SESSION 5: STATISTICAL APPROACHES

Theory and Laboratory – Natural Language Processing (NLP)

List of Recommended Laboratory Programs

The following laboratory programs are arranged progressively, beginning with probability and language models and moving towards statistical parsing and machine-learning-based text classification. Together, they cover the topics listed in the Session 5 syllabus.

Lab No.	Laboratory Program	Description / Expected Implementation
1	**Probability Theory and Language Models**	Implement unigram, bigram and trigram language models. Calculate the probability of a given sentence using word and sequence probabilities.
2	**Discrete-Time Language Models**	Implement an N-gram model to predict the next word in a sentence using the preceding sequence of words.
3	**Markov Models for Text Processing**	Implement a Markov model to calculate state-transition probabilities and predict the next state or word in a sequence.
4	**Entropy and Information Theory in NLP**	Calculate the entropy and cross-entropy of a text corpus using word-probability distributions.
5	**Statistical Parsing Using Probabilistic Context-Free Grammar (PCFG)**	Construct a probabilistic grammar and implement statistical parsing to identify the most probable parse tree for a given sentence.
6	**Text Categorization Using Machine Learning**	Use Bag-of-Words and TF-IDF feature extraction to represent documents and classify them into predefined categories.
7	**Text Classification Using Naive Bayes**	Develop a Naive Bayes classifier to classify text documents, such as spam/ham emails or positive/negative reviews.
8	**Text Classification Using Support Vector Machine (SVM)**	Implement an SVM-based text classifier using TF-IDF features. Evaluate it using accuracy, precision, recall and F1-score.
9	**Text Clustering Using K-Means**	Implement K-Means clustering to group similar text documents into clusters without using predefined category labels.
10	**Centroid-Based Text Classification**	Calculate a centroid for each category using training documents and classify new documents based on their similarity to the category centroids.
11	**Comparative Analysis of Text Classification Algorithms**	Compare Naive Bayes, SVM and centroid-based classification using the same text dataset and evaluate their performance.

Suggested Datasets

Dataset	Suitable Programs
SMS Spam Collection	6, 7, 8, 10, 11
20 Newsgroups	6, 8, 9, 10, 11
IMDB Movie Reviews	6, 7, 8, 9, 10
BBC News Dataset	6, 8, 9, 10, 11
Custom Text Corpus	1, 2, 3, 4, 5

Recommended Python Libraries

Library	Purpose
NLTK	Language models, tokenization, PCFG and entropy.
NumPy	Probability calculations and mathematical operations.
scikit-learn	Naive Bayes, SVM, K-Means and TF-IDF.
spaCy	Natural language preprocessing.
Matplotlib / Seaborn	Performance visualization and plots.

Suggested Laboratory Manual Structure

Part A – Statistical Language Processing

• Experiment 1: Probability and N-gram Language Models

• Experiment 2: Discrete-Time Models

• Experiment 3: Markov Models

• Experiment 4: Entropy Models

• Experiment 5: Statistical Parsing (PCFG)

Part B – Statistical Text Classification

• Experiment 6: Text Categorization

• Experiment 7: Naive Bayes Classification

• Experiment 8: SVM-Based Text Classification

• Experiment 9: Text Clustering Using K-Means

• Experiment 10: Centroid-Based Classification

• Experiment 11: Comparative Performance Analysis

Recommendation for Laboratory Execution

For a four-hour laboratory component, retain all 11 experiments in the laboratory manual. Prioritize Experiments 1, 3, 4, 5, 7, 8, 9 and 10 for hands-on execution. This selection provides practical exposure to statistical NLP, probabilistic parsing and machine-learning-based text processing while covering the syllabus topics.